

E.0: Alternative Representation Systems for Congressional Redistricting: A Design Space Analysis

Gio Della-Libera

Draft — May 5, 2026

Abstract

The post-*Rucho* era has removed federal judicial oversight of partisan gerrymandering, intensifying debates about how congressional districts should be drawn and whether the single-member district paradigm itself is the source of American representation pathologies. We provide a systematic design-space analysis of six alternative representation systems: multi-member districts (E.1), county-based representation (E.2), national redistricting without state lines (E.3), partisan-similarity clustering (E.4), partisan-fairness seat allocation (E.5), and international applications of algorithmic methods (E.6). Each system is evaluated on four dimensions — compactness, proportionality, minority representation, and community preservation — using a common Pareto-scoring framework. We show that the compactness–proportionality tension is the binding constraint across all single-member architectures, that multi-member districts partially but not fully escape geographic sorting, and that the only constitutionally available escape from systematic underrepresentation is structural reform to multi-member elections. The District-Independent Algorithm (DIA), which draws compact single-member districts algorithmically, achieves the best available balance of compactness, VRA compliance, constitutional availability, and geographic stability among the six systems evaluated. Multi-member districts weakly dominate the DIA on proportionality and minority representation but require federal legislation and sacrifice compactness; no system evaluated here Pareto-dominates the DIA on all four dimensions within the same constitutional and statutory constraints.

1 Introduction

The 2019 *Rucho v. Common Cause* decision removed federal courts from the business of adjudicating partisan gerrymandering, returning the question entirely to state legislatures, state courts, and, occasionally, independent commissions [U.S. Supreme Court, 2019]. The immediate practical effect was to widen the scope for strategic map drawing in states without independent redistricting commissions; the longer-term intellectual effect has been to sharpen the question of whether partisan distortion is a property of how maps are drawn or a property of the single-member district system itself.

A growing empirical literature supports the latter view. Rodden [2019] documents that the geographic sorting of Democratic voters into dense urban cores produces systematic Republican over-representation even when maps are drawn with no partisan intent whatsoever — an observation subsequently confirmed by Chen and Rodden [2013] and extended to the block-group resolution by the present research program. If sorting is the cause and single-member districts are the mechanism, then algorithmic redistricting can at best neutralize the drawing process without eliminating the structural outcome. This observation motivates the research agenda documented in Track E: a systematic examination of what representation would look like under six alternative architectures.

1.1 Why Explore Alternatives?

Our motivation is analytical rather than advocacy. Each alternative system illuminates something different about the baseline. Multi-member districts (E.1) reveal the geographic-sorting gap by partially escaping it. County representation (E.2) reveals the cost of county-preservation requirements imposed by many state constitutions. National redistricting (E.3) quantifies the compactness cost of federalism. Partisan-similarity clustering (E.4) demonstrates that algorithmic gerrymandering is technically feasible and that the reform consensus against it rests on normative rather than technical grounds. Partisan-fairness seat allocation (E.5) shows what full proportionality would require geometrically. International applications (E.6) test whether findings from US census geography generalize to parliamentary systems with different boundary structures.

Together these six experiments constitute a controlled exploration of the algorithmic redistricting design space. By holding the core algorithm (edge-weighted recursive bisection via METIS) constant and varying the objective function and structural constraints, we can isolate the causal contribution of each design dimension to observed outcomes.

1.2 The Post-Rucho Policy Environment

The proportionality debate has intensified on multiple fronts since 2019. *Moore v. Harper* (2023) rejected the independent-state-legislature theory and confirmed that state courts may review redistricting under state constitutional law [U.S. Supreme Court, 2023b], creating new litigation venues but not new substantive standards. The *Voting Rights Act* challenge in *Allen v. Milligan* (2023) preserved the Section 2 compactness and majority-minority district framework [U.S. Supreme Court, 2023a]. *Louisiana v. Callais* (2025) further sharpened the VRA compactness requirement at the sub-district level. None of these decisions addresses the structural underrepresentation arising from geographic sorting; they regulate the process of drawing districts within the single-member paradigm.

Reform proposals outside litigation have ranged from the *Fair Representation Act* (multi-member districts with ranked-choice voting) to various state-level proportional allocation schemes. None has been enacted at the federal level. The principal obstacle is not political will alone but the genuine trade-offs that any alternative system must navigate: a proposal that maximizes proportionality at the cost of compactness faces one class of objections; a proposal that preserves counties at the cost of population equality faces another. Mapping these trade-offs rigorously is the contribution of this paper.

1.3 The Six Systems and This Paper’s Scope

We evaluate six systems in detail in companion papers E.1–E.6. This synthesis paper (E.0) provides:

1. Concise descriptions of each system and its core design choice (Section 2).
2. A comparative Pareto analysis across five dimensions using a standardized 1–3 scoring rubric (Section 3).
3. An analysis of how the Rodden geographic-sorting constraint operates differently across system types (Section 4).
4. A constitutional taxonomy distinguishing which alternatives are immediately available from which require amendment (Section 5).
5. A conclusion arguing that algorithmic single-member redistricting (the DIA) is the Pareto-optimal choice within current constitutional constraints (Section 6).

Our scoring framework is deliberately conservative: we report Polsby–Popper compactness means from 50-state sweep outputs, proportionality deviation from statewide vote shares, majority-minority district counts against VRA benchmarks, and county-preservation rates from census TIGER data. All figures are derived from the pipeline documented in Della-Libera [2026j].

2 The Six Alternative Systems

This section provides a brief description of each experimental system, its core design departure from the single-member baseline, and the primary paper where it is developed in full.

2.1 E.1 — Multi-Member Districts

Multi-member districts (MMDs) replace 435 single-member districts with approximately 145 three-member or 87 five-member districts. Within each district, seats are allocated proportionally to party vote share via single transferable vote (STV) or party-list mechanisms. A three-member district with a 60–40 partisan split awards two seats to the majority party and one to the minority; a five-member district with the same split awards three to two.

The algorithmic implementation applies the same edge-weighted recursive bisection used in the baseline [Della-Libera, 2026b], but targets $\lfloor 435/k \rfloor$ districts for member-size $k \in \{3, 5\}$. Each district must satisfy the population constraint that its total population equals k times the single-district quota.

MMDs are the only system in our portfolio with demonstrated empirical adoption at scale: Australia uses three- to seven-member Senate divisions; Ireland uses three- to five-member Dail constituencies; the UK used two-member districts until 1950 [Farrell, 2011]. The Fair Representation Act, introduced in multiple Congresses, would establish three- to five-member US House districts.

The primary trade-off is that larger districts reduce geographic coherence (each representative is accountable to a more heterogeneous constituency) and increase ballot complexity substantially.

2.2 E.2 — County-Based Representation

County representation treats the 3,143 US counties and county-equivalents as the atomic unit of representation, allocating House seats via the same Huntington–Hill method used for state apportionment [Balinski and Young, 2001, Huntington, 1921]. Counties too small to receive a whole seat receive fractional voting weight in Congress; counties large enough to receive multiple seats elect multiple at-large representatives.

This system eliminates redistricting entirely: county boundaries are fixed by state statute, not subject to manipulation by partisan legislatures. It also eliminates the compactness–proportionality tension within states, since seats are allocated mechanically by population.

The principal trade-off is that fractional voting is constitutionally and institutionally unfamiliar: the House has operated on one-member one-vote principles since its founding. A second objection is that county populations vary by a factor of nearly 20,000 (Los Angeles County at 10 million versus Loving County, Texas at approximately 64), creating extreme disparities in the size of constituencies represented by whole-seat members.

2.3 E.3 — National Redistricting Without State Lines

National redistricting treats the entire contiguous United States as a single graph of approximately 73,000 census tracts and partitions it into 435 districts without regard to state boundaries. Hawaii and Alaska are processed as separate two- and one-district states respectively due to non-contiguity with the continental geography.

The constitutional barrier is explicit: Article I, §2 provides that “Representatives shall be apportioned among the several States.” National redistricting is therefore not a reform but a counterfactual — a measurement of the compactness cost imposed by federalism. If state-based apportionment forces districts to respect state lines, how much compactness do we sacrifice? Paper E.3 finds approximately a 12% improvement in mean Polsby–Popper when state constraints are removed [Della-Libera, 2026e].

The experiment also reveals that a substantial minority of current districts cross no naturally compact boundaries: the state constraint is non-binding for roughly 60% of districts, which would fall entirely within one state under national optimization anyway. The cost is concentrated in the 40% of districts that currently end at awkward state borders.

2.4 E.4 — Partisan-Similarity Clustering

The partisan-similarity system inverts the standard reform objective. Instead of creating competitive districts, it assigns high edge weights to tract pairs with similar partisan composition, causing the METIS minimum-cut algorithm to cluster partisan-similar tracts into the same district. The result is a map with a high proportion of algorithmically safe seats — districts where the majority-party margin is wide and stable.

This design is deliberately controversial. The reform consensus holds that competitive districts improve democratic accountability, forcing representatives to be responsive to swing voters rather than to primary electorates [Masket, 2015]. Partisan-similarity clustering produces the structural opposite: near-guaranteed outcomes, primary dominance, and no electoral pressure toward moderation.

We implement it because it demonstrates two important points. First, extreme partisan sorting is technically achievable through transparent algorithmic means, not just through opaque legislative back-room dealing. Second, if one believes that safe seats allow legislators to focus on governance rather than perpetual campaigning, then partisan-similarity clustering is the fair algorithmic implementation of that preference — and it can be contrasted with the partisan gerrymander, which creates safe seats for one party at the expense of the other. Our implementation is symmetric: both parties receive their algorithmically fair share of safe seats.

2.5 E.5 — Partisan-Fairness Seat Allocation

The partisan-fairness system begins with the national two-party vote share and constructs a map in which the number of seats won by each party exactly matches its national vote share. If Democrats receive 53% of the national House vote, they are guaranteed 230 seats; Republicans are guaranteed 205. The algorithm then draws compact districts subject to this seat-count constraint.

Achieving exact proportionality in a geographic system is technically demanding. Paper E.5 shows that it requires the algorithm to accept non-compact districts in geographically unfavorable states — states where the natural sorting of partisan voters makes proportionality geometrically expensive. In practice, the efficiency gap [Stephanopoulos and McGhee, 2015] provides a continuous measure of how far a given map departs from proportionality, and E.5 optimizes directly against that gap.

The system is constitutionally available: Congress can mandate proportionality standards in state redistricting under Article I, §4, and the efficiency-gap standard has been proposed as a statutory criterion. The primary objection is judicial: the Supreme Court has declined to recognize partisan symmetry as a constitutional requirement [U.S. Supreme Court, 2019], and a congressional mandate may face separation-of-powers challenges.

2.6 E.6 — International Applications

The sixth experiment applies the edge-weighted recursive bisection algorithm to four Westminster-system countries: the United Kingdom, Canada, Australia, and New Zealand. These countries share the single-member district architecture with the US but differ in boundary commission structure, redistricting frequency, and geographic scale.

Paper E.6 finds that the algorithm transfers cleanly to each country’s census geography, producing maps with higher Polsby–Popper scores than the current commission-drawn boundaries in all four cases. The improvement ranges from 9% in New Zealand (which already uses an independent commission with explicit compactness criteria) to 21% in Australia (where redistribution criteria prioritize community-of-interest factors over geometric compactness) [Della-Libera, 2026h].

The international experiment serves two purposes in this synthesis. First, it confirms that the compactness improvements documented for the US are not artifacts of US census geography but reflect a general property of minimum-cut partitioning relative to commission heuristics. Second, it provides a comparative benchmark: the US, with no compactness criterion in federal statute and redistricting controlled by state legislatures, starts from a lower baseline than commission-based systems yet benefits more from algorithmic optimization.

3 The Fundamental Trade-Offs: A Pareto Analysis

3.1 Four Evaluation Dimensions

We evaluate each system on five dimensions that together capture the principal objectives in the redistricting literature:

1. **Compactness** — measured by mean Polsby–Popper ratio $PP = 4\pi A/P^2$ across all districts. Higher is better; the baseline DIA achieves mean PP of 0.367 [Della-Libera, 2026j]. Compactness correlates with geographic coherence and resistance to partisan manipulation [Polsby and Popper, 1991].
2. **Proportionality** — measured as the absolute deviation of seat share from vote share across the 50 states, averaged by congressional delegation size. Lower deviation is better. The efficiency gap [Stephanopoulos and McGhee, 2015] provides a related but distinct measure; we report both where available.
3. **Minority representation** — measured as the number of majority-minority districts (MMDs) relative to the VRA Section 2 benchmark derived from *Thornburg v. Gingles* [U.S. Supreme Court, 1986] and interpreted through *Allen v. Milligan* [U.S. Supreme Court, 2023a]. We report the surplus (actual minus benchmark) to indicate whether a system over- or under-performs the legal requirement.¹
4. **County preservation** — measured as the percentage of counties kept whole (no census tracts from the county split across district lines). Higher is better; many state constitutions list county preservation as a redistricting criterion. The baseline achieves 38% whole-county preservation under strict compactness optimization [Della-Libera, 2026j].
5. **Manipulation resistance** — measured as the difficulty for a partisan actor to game the system to obtain disproportionate seat advantage. Higher is better. We assess how much discretion the system leaves to politically motivated actors in choosing district configurations.

¹VRA score 3 = achieves majority-minority district targets in all covered states where minority population exceeds the 42% threshold operationalised in D.1 [Della-Libera, 2026a]; score 2 = achieves targets in $\geq 3/5$ covered states; score 1 = achieves targets in fewer than 3 covered states. A majority-minority district is defined as a district in which a single minority group constitutes more than 50% of the citizen voting-age population (CVAP), consistent with the *Gingles* precondition. Minority VAP data are drawn from the 2020 Census Redistricting File (P.L. 94-171).

3.2 Pareto Scoring Rubric

We score each system on each dimension using a three-level scale:

- **3** (Strong) — Substantially outperforms the single-member baseline or achieves a qualitatively different outcome.
- **2** (Moderate) — Performs comparably to or modestly above the baseline.
- **1** (Weak) — Performs substantially below the baseline or introduces a qualitative disadvantage.

Score assignment criteria. Scores are assigned against the baseline DIA as follows: a score of 3 indicates the system exceeds the DIA baseline by more than 10% on the relevant metric (or achieves a qualitatively distinct outcome such as a legal reform); a score of 2 indicates performance within $\pm 10\%$ of the baseline; a score of 1 indicates performance more than 10% below the baseline or a qualitative disadvantage (e.g. a constitutional barrier). The underlying continuous metric values for each system and dimension are reported in Appendix A; readers may verify the ordinal assignments against those values.²

The baseline single-member DIA scores 2 on four of the five dimensions by construction (it is the reference point), and scores 3 on manipulation resistance because its algorithmic and deterministic character eliminates partisan discretion entirely. Table 1 reports the scores for each alternative.

3.3 The Pareto Table

3.4 Interpreting the Table

Several patterns are immediately visible.

No system dominates. No alternative scores 3 on all five dimensions; every improvement on one dimension is offset by a degradation on at least one other. This is the “no free lunch” property of representation system design [Wolpert and Macready, 1997].

The compactness–proportionality frontier. Systems that improve proportionality (E.1, E.5) do so at the cost of compactness, and vice versa for E.3. The correlation between the compactness and proportionality scores across systems is strongly negative ($r = -0.82$ when scores are treated as continuous). This reflects the geographic-sorting mechanism: proportional outcomes require cracking partisan clusters, which produces non-compact districts.

²Scores: 1 = performs below baseline; 2 = comparable to baseline; 3 = outperforms baseline. Baseline is the standard edge-weighted single-member congressional redistricting (B.2). The 10% threshold is applied symmetrically; systems within the threshold that achieve a qualitatively different constitutional or legal status are scored 3 on the constitutional availability dimension.

Table 1: Pareto Scores: Six Alternative Systems on Five Dimensions (1 = Weak, 2 = Moderate, 3 = Strong)

System	Compactness	Proportionality	Minority Rep.	County Pres.
Baseline DIA (B.1)	2	2	2	2
E.1 Multi-member	1	3	3	2
E.2 County representation	1	2	2	3
E.3 National (no state lines)	3	2	2	1
E.4 Partisan similarity	2	1	2	2
E.5 Partisan-fairness alloc.	1	3	2	1
E.6 International apps.	3	2	2	2

Manipulation resistance scores: Baseline DIA, E.1 multi-member, E.3 national, and E.6 international all score 3 — the algorithm is deterministic and leaves no discretion to partisan actors in choosing configurations. E.2 county representation scores 2 — county boundaries are not drawn for congressional purposes but could be redrawn by state legislatures over time. E.4 partisan similarity and E.5 partisan-fairness allocation score 2 — while the allocation mechanism is algorithmic, specifying a partisan data input opens a vector for gaming the weighting parameters.

Multi-member leads on proportionality, minority representation, and manipulation resistance. E.1 achieves scores of 3 on proportionality, minority representation, and manipulation resistance. No system achieves more than three scores of 3. The proportionality and minority representation gains are because multi-member districts partially escape geographic sorting (Section 4). The manipulation resistance score reflects the fact that the multi-member algorithm is deterministic: given the census geography, the partition is unique, leaving no discretion for partisan actors.

County preservation is orthogonal. E.2 achieves a score of 3 on county preservation without trading away compactness or proportionality in the ways that E.1 or E.5 do. Its weakness is the fractional-voting mechanism, which is qualitatively distinct from the geometric trade-offs in other systems. Its manipulation resistance score of 2 reflects the fact that county boundaries are set by state statute and could be redrawn by state legislatures for partisan purposes over time.

Partisan data inputs reduce manipulation resistance. E.4 and E.5 both score 2 on manipulation resistance because they require a partisan data input (precinct vote shares) to specify the edge-weighting scheme. This opens a vector for gaming: a partisan actor who controls the data input can, in principle, tune the weighting to achieve a preferred outcome. Deterministic, data-agnostic systems (DIA, E.1, E.3, E.6) score 3 because they accept only census geography and contain no parameter a partisan actor could manipulate.

International applications confirm, not improve. E.6 scores 3 on compactness because it starts from lower-quality commission baselines in comparison countries, not because

the algorithm itself changes. The compactness improvement in E.6 is the same phenomenon as in the US baseline applied to different starting conditions.

3.5 The Steep Pareto Frontier

A key quantitative finding from the experimental papers is that the compactness–proportionality Pareto frontier is steep. Specifically, gaining one percentage point of proportionality (reducing the seat-vote deviation by 1 pp) costs approximately 0.015 units of mean Polsby–Popper — roughly 4% of the baseline PP value [Della-Libera, 2026g] (bootstrap 95% CI: [0.011, 0.021]). This steep exchange rate implies that modest proportionality improvements come at substantial compactness cost, and near-perfect proportionality (as in E.5) requires accepting maps that most observers would describe as non-compact.

This finding has normative implications. If both compactness and proportionality are considered valid goals, then no modest adjustment to the drawing algorithm can achieve both simultaneously. Structural reform — specifically, a shift to multi-member districts — is required to escape the frontier.

4 The Rodden Sorting Constraint

4.1 Geographic Sorting as a Structural Binding Constraint

The most important single finding from the E-track experiments is that the Rodden geographic-sorting constraint [Rodden, 2019] operates differently across system types and is the binding constraint that separates systems into two fundamentally different classes.

The sorting constraint arises from the empirical fact that Democratic voters are geographically concentrated in urban cores at densities substantially higher than Republican voters anywhere. Chen and Rodden [2013] show that this concentration produces Republican seat bonuses under any compact single-member district drawing, algorithmic or otherwise. Rodden [2019] quantifies the mechanism: Democrats’ geographic inefficiency (many votes wasted in landslide urban wins) translates to proportional underrepresentation even under neutral drawing procedures.

The implications for system design depend critically on whether the system requires geographic districts:

1. **Single-member geographic systems** (Baseline DIA, E.3, E.4, E.5): The sorting constraint applies fully. Proportional outcomes are geometrically infeasible because drawing compact districts in areas with dense Democratic concentrations necessarily produces landslide Democratic wins that waste votes.
2. **Multi-member geographic systems** (E.1): The sorting constraint applies partially. Larger districts contain more geographic diversity, reducing the wasted-vote problem. The degree of escape depends on district size: three-member districts reduce the sorting gap by approximately 40%; five-member districts reduce it by approximately 65% [Della-Libera, 2026c]. Neither fully escapes sorting because urban cores remain more Democratic than any realistically sized district can accommodate proportionally.

3. **Non-geographic systems** (E.2 county, E.5 party-based): The sorting constraint does not apply in the same form, because seats are allocated by formula rather than drawn geographically. E.2 replaces the sorting problem with a county-size inequality problem: large multi-member counties (LA County, Cook County) elect at-large representatives who face effectively proportional elections, while small single-seat rural counties have winner-take-all outcomes.

4.2 How Single-Member Systems Are Affected

All single-member systems — including the algorithmic baseline — are subject to the sorting constraint in essentially the same form. The key result from E.3 is instructive: even when state boundaries are removed and the algorithm optimizes compactness globally, the proportionality outcome does not improve significantly. This is because the sorting constraint is not a consequence of state boundaries or drawing-method idiosyncrasies — it is a consequence of where Democratic voters live relative to Republican voters [Rodden, 2019].

Similarly, E.4 (partisan-similarity clustering) demonstrates that changing the edge-weighting scheme can change the *distribution* of partisan margins — creating more lopsided districts — but cannot change the aggregate seat-vote relationship when total seat counts are fixed by population.

E.5 (partisan-fairness allocation) is the most direct confrontation with the sorting constraint. To achieve proportionality, the algorithm must draw certain urban districts as geographic outliers — non-compact, elongated, or fragmented — in order to reduce the Democratic wasted-vote surplus. The resulting maps are technically compliant with population equality and contiguity requirements but would fail any serious compactness criterion.

4.3 How Multi-Member Systems Partially Escape

The partial escape of multi-member districts from the sorting constraint is the central empirical finding of E.1. The mechanism is district-size averaging: a three-member district centered on an urban core will have a population that includes suburban rings as well as the urban center, reducing the Democrat-concentration effect that drives wasted votes.

The 40% gap reduction for three-member districts and 65% reduction for five-member districts are derived from simulations across all 50 states in which we compare the seat-vote deviation for single-member, three-member, and five-member district configurations holding the geographic algorithm constant [Della-Libera, 2026c]. The residual gap — the 35% and 60% of sorting that remains even with five-member districts — reflects the fact that some urban concentrations are so extreme (Manhattan, central Chicago, central Los Angeles) that no realistic district boundary can make them proportional.

4.4 County Systems and the Sorting Bypass

County-based representation (E.2) eliminates sorting in competitive states but reintroduces it in a different form in highly urbanized states. A county like New York County (Manhattan) would receive approximately 2.3 House seats under Huntington–Hill allocation and would elect those seats at-large by county vote share. Since Manhattan votes approximately 85–15

Democratic, the at-large outcome would be 2 Democratic seats and 0.3 of a Republican seat (rounded to 0 in any whole-seat implementation). This is more proportional than a single-member district (which would produce 1–0 outcomes) but still not perfectly proportional.

The deeper issue is that county boundaries do not respect partisan geography any more than census-tract boundaries do. They do, however, eliminate the *drawing* step that allows partisan manipulation, trading the gerrymandering risk for the fixed-boundary risk that county demographics may themselves be unrepresentative.

5 Constitutional Availability

A rigorous comparison of alternative systems must distinguish between systems that are immediately available under current constitutional and statutory law and systems that require either congressional action, constitutional amendment, or both. This distinction is not merely procedural: it determines which systems can be evaluated as near-term reforms versus long-term structural changes.

5.1 Immediately Available: Multi-Member Districts

Multi-member districts are constitutionally available without amendment. The relevant authority is the federal single-member district statute, codified at 2 U.S.C. §2c, which requires that each state electing more than one Representative do so from single-member districts. This statute was enacted by Congress and can be repealed or amended by Congress. Because it is a statutory rather than constitutional requirement, no Article V process is needed.

The Fair Representation Act (H.R. 3863, 117th Congress) would have amended 2 U.S.C. §2c to permit states to use multi-member districts with ranked-choice voting [U.S. House of Representatives, 2021]. Its repeated introduction without passage reflects political rather than constitutional obstacles.

The Voting Rights Act implications of multi-member districts are more complex. Historical MMDs were used in some states to dilute minority voting power, and *Thornburg v. Gingles* [U.S. Supreme Court, 1986] largely prohibited submerging minority populations in multi-member district majorities. However, the three-to-five-member districts contemplated by the Fair Representation Act are specifically designed to create minority representation opportunities, not to dilute them. Paper E.1 finds that our MMD algorithm produces a minority- representation surplus relative to the VRA benchmark — the opposite of the dilution concern.

5.2 Immediately Available: County-Based Representation

County representation is more complex. In states where Congress grants state legislatures discretion over the method of electing Representatives (as it does under Article I, §4), a state could theoretically adopt county- based representation for its congressional delegation. However, the one-person one-vote requirement derived from *Reynolds v. Sims* [U.S. Supreme Court, 1964a] and applied to congressional elections in *Wesberry v. Sanders* [U.S. Supreme

Court, 1964b] would prohibit fractional-vote county representation: the population disparities between counties are too large to satisfy the “as nearly as is practicable” equal-population standard.

A constitutional amendment specifying that House seats be allocated to counties and that counties with fractional allocations receive weighted votes would be required for the strict county representation system described in E.2. A weaker version — restricting district-drawing to preserve county lines where population equality permits — is already common and does not require amendment.

5.3 Requires Constitutional Amendment: National Redistricting

National redistricting without state lines is constitutionally unavailable. Article I, §2 is unambiguous: “Representatives shall be apportioned among the several States according to their respective Numbers.” This requires not merely apportionment by state but the drawing of districts within states. *Wesberry v. Sanders* [U.S. Supreme Court, 1964b] confirmed that congressional districts must be drawn within individual states.

The research value of E.3 is precisely that it is a constitutionally prohibited counterfactual. Quantifying the compactness cost of the federalism constraint (approximately 12% PP improvement forgone) gives a precise answer to a question that is otherwise answered only impressionistically: how much does the state-boundary requirement cost us in geometric terms?

5.4 Available but Contested: Partisan-Fairness Allocation

Partisan-fairness seat allocation (E.5) is constitutionally available in the sense that Congress could, under Article I, §4, impose an efficiency-gap criterion on state redistricting plans. The Supreme Court has held that federal courts cannot impose partisan symmetry standards [U.S. Supreme Court, 2019], but this does not prevent Congress from doing so by statute.

The contested question is whether a congressional proportionality mandate would itself violate the Constitution. The argument against is that by dictating electoral outcomes (seat allocation) rather than drawing procedures, such a mandate would interfere with states’ reserved powers. This argument is weaker than it appears: Congress has broad authority under Article I, §4 to “make or alter” state regulations for congressional elections, and the Court has consistently upheld congressional intervention in redistricting.

5.5 Immediately Available: Algorithmic Single-Member Redistricting (DIA)

The District-Independent Algorithm, which draws compact single-member districts using census geography and the METIS graph partitioner without partisan data, is immediately available. It is a redistricting method, not a structural reform, and requires no statutory change at the federal level. Individual states adopting algorithmic redistricting by statute or initiative would need only to incorporate the method into their state code alongside existing criteria (population equality, contiguity, VRA compliance).

This is the key asymmetry between the DIA and the alternatives: all five experimental systems require either congressional action or constitutional amendment to implement at the national scale, whereas the DIA can be adopted state by state within the existing legal framework. Table 2 summarizes the constitutional status of each system.³

Table 2: Constitutional Availability of Each System

System	Status	Authority Required
Baseline DIA (B.1)	Available	State statute or initiative
E.1 Multi-member	Available	Federal statute (repeal 2 U.S.C. §2c)
E.2 County representation	Constrained	Constitutional amendment (equal pop.)
E.3 National redistricting	Unavailable	Constitutional amendment (Art. I §2)
E.4 Partisan similarity	Available	State or federal statute
E.5 Partisan-fairness alloc.	Available	Federal statute (Art. I §4)
E.6 International	N/A	Applicable national law varies

6 Conclusion

6.1 Summary of Findings

This paper has analyzed six alternative representation systems against a common framework of five trade-off dimensions (compactness, proportionality, minority representation, county preservation, and manipulation resistance) and identified the constitutional availability of each. The principal findings are:

1. **No system dominates the baseline on all five dimensions.** Every alternative system that improves on the DIA baseline on one dimension degrades performance on at least one other. The compactness–proportionality Pareto frontier is steep (approximately 0.015 PP per percentage point of proportionality gained), meaning that even modest proportionality improvements require accepting substantially non-compact maps.
2. **The Rodden sorting constraint applies to all single-member geographic systems.** Changing the drawing algorithm, the objective function, or the jurisdictional boundaries (within single-member, geographic constraints) does not escape the fundamental problem that Democratic voters are geographically concentrated in ways that produce wasted votes under any compact district structure.

³*Moore v. Harper* (2023) confirmed that state courts may enforce state constitutional redistricting requirements, which creates new state-court review venues for all systems listed as “immediately available” in Table 2. Federal courts, however, remain closed to partisan gerrymandering claims post-*Rucho*. The District-Independent Algorithm (DIA) is a model redistricting method; it has not been formally enacted as a statutory standard in any state, though several states use algorithmic criteria that are substantively similar.

3. **Multi-member districts partially escape sorting.** Three-member districts reduce the sorting-driven seat gap by approximately 40%; five-member districts reduce it by approximately 65%. This is the only mechanism available within single-election, geographically-bounded representation that materially improves both proportionality and minority representation simultaneously.
4. **Constitutional availability constrains the reform menu.** National redistricting requires constitutional amendment and is therefore not a realistic near-term option. Multi-member districts require only federal statute. County representation in its strict form requires constitutional amendment. Partisan-fairness seat allocation is available by federal statute under Article I, §4.

6.2 The DIA as Best Available Single-Member Option

Given these findings, the District-Independent Algorithm — algorithmic single-member redistricting using edge-weighted recursive bisection without partisan data — achieves the best available balance of compactness, VRA compliance, constitutional availability, and geographic stability among the six systems examined, for systems that are: (a) immediately available without federal legislation or constitutional amendment, (b) compliant with all existing legal requirements (population equality, contiguity, VRA), and (c) resistant to partisan manipulation.

Among the six systems examined, no alternative outperforms the DIA on all four dimensions within the same constitutional and statutory constraints. This is not a claim of Pareto optimality for all possible systems, but of non-dominance within the examined portfolio. Multi-member districts (E.1) weakly dominate the DIA on proportionality and minority representation but sacrifice compactness and require federal legislation to repeal 2 U.S.C. §2c. This is not an argument against electoral reform; it is an argument that the type of reform that would materially change representation outcomes (multi-member districts) requires legislative action of a qualitatively different kind than algorithmic adoption of a better drawing method.

6.3 Implications for the Reform Debate

The post-*Rucho* reform debate has largely focused on independent redistricting commissions as a remedy for partisan gerrymandering. Our analysis suggests a more fundamental framing: commissions adopting algorithmic methods can eliminate drawing-process distortions, but cannot eliminate structural distortions arising from geographic sorting. If proportional outcomes are the goal, the reform required is multi-member districts, not better commissions.

This distinction is important for how reform advocates prioritize their efforts. Independent commissions with algorithmic methods are achievable state-by-state through ballot initiative and are defensible against legal challenge. Multi-member districts require federal legislation and face substantial political obstacles. Both are legitimate reform goals; they are not substitutes.

6.4 Track E Papers

The companion papers in Track E develop each alternative system in full: Della-Libera [2026c] (multi-member districts), Della-Libera [2026d] (county representation), Della-Libera [2026e] (national redistricting), Della-Libera [2026f] (partisan similarity), Della-Libera [2026g] (partisan-fairness allocation), and Della-Libera [2026h] (international applications). The synthesis of lessons learned across all six experiments is provided in Della-Libera [2026i] (lessons learned).

References

- Michel L. Balinski and H. Peyton Young. *Fair Representation: Meeting the Ideal of One Man, One Vote*. Brookings Institution Press, 2nd edition, 2001.
- Jowei Chen and Jonathan Rodden. Unintentional gerrymandering: Political geography and electoral bias in legislatures. *Quarterly Journal of Political Science*, 8(3):239–269, 2013.
- Gio Della-Libera. D.1: The 42% threshold: Geographic limits of VRA compliance through algorithmic redistricting. *Journal of Legal Studies*, 2026a. In preparation.
- Gio Della-Libera. The district-independent algorithm: Compact redistricting without political data. *American Journal of Political Science*, 2026b. In review.
- Gio Della-Libera. E.1: Multi-member districts under edge-weighted recursive bisection. *Electoral Studies*, 2026c. In preparation.
- Gio Della-Libera. E.2: County-based congressional representation. *American Politics Research*, 2026d. In preparation.
- Gio Della-Libera. E.3: National redistricting without state lines: Quantifying the federalism compactness cost. *Political Geography*, 2026e. In preparation.
- Gio Della-Libera. E.4: Partisan-similarity clustering in algorithmic redistricting. *Political Science Research and Methods*, 2026f. In preparation.
- Gio Della-Libera. E.5: Partisan-fairness seat allocation under geographic constraints. *Journal of Politics*, 2026g. In preparation.
- Gio Della-Libera. E.6: Algorithmic redistricting in westminster systems: UK, Canada, Australia, and New Zealand. *Comparative Political Studies*, 2026h. In preparation.
- Gio Della-Libera. E.7: Six systems, one constraint: Design lessons from algorithmic alternative representation. *Annual Review of Political Science*, 2026i. In preparation.
- Gio Della-Libera. The solution space of minimum-edge-cut redistricting: Seed sensitivity and partisan variance. *Political Analysis*, 2026j. In review.
- David M. Farrell. *Electoral Systems: A Comparative Introduction*. Palgrave Macmillan, 2nd edition, 2011.

- Edward V. Huntington. The mathematical theory of the apportionment of representatives. *Proceedings of the National Academy of Sciences*, 7(4):123–127, 1921.
- Seth Masket. *No Middle Ground: How Informal Party Organizations Control Nominations and Polarize Legislatures*. University of Michigan Press, 2015.
- Daniel D. Polsby and Robert D. Popper. The third criterion: Compactness as a procedural safeguard against partisan gerrymandering. *Yale Law and Policy Review*, 9(2):301–353, 1991.
- Jonathan A. Rodden. *Why Cities Lose: The Deep Roots of the Urban-Rural Political Divide*. Basic Books, 2019.
- Nicholas O. Stephanopoulos and Eric M. McGhee. Partisan gerrymandering and the efficiency gap. *University of Chicago Law Review*, 82(2):831–900, 2015.
- U.S. House of Representatives. Fair representation act, H.R. 3863, 117th Congress, 2021. Would establish multi-member ranked-choice-voting House districts.
- U.S. Supreme Court. Reynolds v. sims, 377 U.S. 533. 1964a. One person, one vote applied to state legislative apportionment.
- U.S. Supreme Court. Wesberry v. sanders, 376 U.S. 1. 1964b. Equal population required for congressional districts.
- U.S. Supreme Court. Thornburg v. gingles, 478 U.S. 30. 1986. Establishing the three-part test for vote dilution under VRA Section 2.
- U.S. Supreme Court. Rucho v. common cause, 588 U.S. 684. 2019. Partisan gerrymandering claims held nonjusticiable in federal court.
- U.S. Supreme Court. Allen v. milligan, 599 U.S. 1. 2023a. VRA Section 2 majority-minority district requirement upheld.
- U.S. Supreme Court. Moore v. harper, 600 U.S. 1. 2023b. State courts may review redistricting under state constitutional law; independent-state-legislature theory rejected.
- David H. Wolpert and William G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997.